

L9b: Pricing American Style Options Contracts using the Cox, Ross, Rubinstein (CRR) Binomial Model

CHEME 5660 Cornell University

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Objectives for Today

Previously, we priced European options, which can be exercised only at expiration. Standard options on stocks and ETFs are American-style and can be exercised at any time. What is the right to exercise early worth?

Today's concepts

- Distinguish between European and American options: exercise only at expiration, or at any time before expiration.
- Apply the binomial model to price American options: build CRR trees and compute premiums with the decision rule for early exercise.
- Estimate the premium of American options by simulation: Longstaff–Schwartz Monte Carlo, regressing the value of holding on the share price.

Lectures and Examples

Lecture:
CHEME-5660-L9b-Lecture-American-Derivatives-Intrinsic-Extrinsic-Value-Fall-2026.ipynb

Example 1:
CHEME-5660-L9b-Example-AmericanContractsPremium-Fall-2026.ipynb

Example 2:
CHEME-5660-L9b-Example-LSM-Premium-Fall-2026.ipynb

Review on your own:
CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb

Company Profile: The Options Clearing Corporation

OCC stands between the buyer and the seller of every U.S. listed option. It opened in 1973 as CBOE Clearing Corporation and became OCC in 1975.

- Exercise and assignment: OCC assigns each early exercise at random to a clearing member with a short position in the same contract.
- Automatic exercise at expiration: since 1977, an expiring equity option in the money by 0.01 USD/share or more is exercised unless the holder instructs otherwise.
- Monte Carlo margin and binomial trees: since 2006, STANS simulates 10,000 two-day scenarios and reprices American options on a binomial tree.

Connection to today's lecture: a seller of an American put can be assigned once exercising now beats holding, so a European-premium margin would leave out the early-exercise premium.

American Style Options Contracts

American-style options contracts allow the holder to exercise the option at any time before or on the expiration date. Previously, we wrote the premium of an American-style contract as:

$$\mathcal{P}^{\text{AM}} = \max_{\tau \leq T} \mathbb{E}_{\mathbb{Q}}\Big(\mathcal{D}_{\tau,0}^{-1}(g_y) \cdot V(K, S(\tau))\Big) \geq \mathcal{P}^{\text{EU}}$$

The exercise time τ is chosen using only the information available at that time (a stopping time). The American premium is at least the European one because the American holder can always wait until expiration.

The standard method to price American options is the Cox, Ross, Rubinstein (CRR) binomial model.

Reference: Cox, J. C., Ross, S. A., and Rubinstein, M. (1979). Option pricing: A simplified approach. Journal of Financial Economics 7(3): 229.

CRR Binomial Lattice Model

At each time step, the share price moves up by a factor u with the risk-neutral probability q, or down by a factor d with probability $(1-q)$. The CRR parameters are no longer data driven: q makes the expected return of the underlying asset match the risk-free rate g_y.

$$
\begin{aligned}
\mathcal{D}_{1,0}(g_y) \cdot S_0 &= \mathbb{E}_{\mathbb{Q}}\left(S_1\right) && \text{grow at the risk-free rate under } \mathbb{Q} \\
&= q \cdot (u \cdot S_0) + (1-q) \cdot (d \cdot S_0) && \text{up with probability } q\text{, down with } (1-q) \\
&= S_0(qu + (1-q)d) && \text{factor out } S_0 \\
&= S_0(qu + d - qd) && \text{expand } (1-q)d \\
&= S_0(d + q(u-d)) && \text{collect the } q \text{ terms} \\
\implies & \boxed{q = \frac{\mathcal{D}_{1,0}\left(g_y\right) - d}{u-d}} && \text{divide by } S_0 \text{ and solve for } q
\end{aligned}
$$

Here $\mathcal{D}_{1,0}(g_y) = e^{g_y \Delta t}$ is the risk-free accumulation factor for one time step. This is the risk-neutral probability we found by ruling out arbitrage when we introduced lattice models, with $R_f = \mathcal{D}_{1,0}(g_y)$.

CRR Up and Down Factors

Unlike the data-driven binomial model we discussed earlier, the CRR model derives u and d based on the volatility σ of the underlying asset, the time step Δt, and the symmetric condition $ud = 1$:

$$\boxed{\begin{aligned} u &= e^{\sigma\sqrt{\Delta t}} \\ d &= e^{-\sigma\sqrt{\Delta t}} \end{aligned}}$$

Here σ is the same volatility as in geometric Brownian motion and the Black–Scholes–Merton formula (units: $\mathrm{yr}^{-1/2}$).

Where do u and d come from? advanced/crr_factors/CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

Historical or Implied Volatility?

There are two ways to put a number on σ, and the second also gives a quick range for the share price:

- Historical volatility: previously, we estimated σ from past share prices as $\sigma_{\text{ann}} = \sqrt{\Delta t} \cdot \sigma_g$, where σ_g is the standard deviation of the growth rates and Δt is the time between price observations. This estimate looks backward.
- Implied volatility: the implied volatility (IV) is the market's forecast of a likely movement in the share price, the volatility "implied" by an option's market price through a pricing model such as Black–Scholes–Merton or CRR. It is often equal to or larger than the historical volatility.
- How can we use it? IV represents a one-standard-deviation movement in the share price over the next year. Over a horizon of T years, the size of that move scales with $\sqrt{T}$.

A Back-of-the-Envelope Range from IV

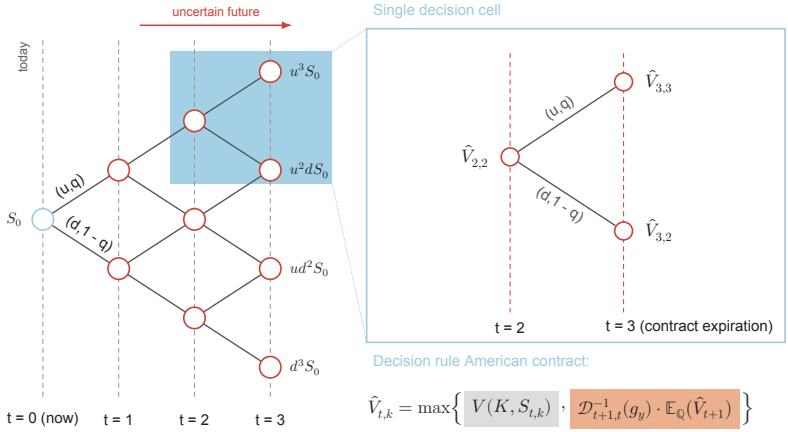
Moving one standard deviation up or down in the solution of geometric Brownian motion gives a back-of-the-envelope range for the share price:

$$
\begin{aligned}
S(T) &= S_0 \exp\left[\mu_g T + \sigma\sqrt{T}\, Z\right] && \text{solution of geometric Brownian motion} \\
S_\pm &= S_0 \exp\left[\mu_g T \pm \text{IV}\sqrt{T}\right] && \text{set } \sigma = \text{IV and } Z = \pm 1 \\
&\approx S_0 \exp\left[\pm \text{IV}\sqrt{T}\right] && \mu_g T \text{ is small next to IV}\sqrt{T} \text{ for short } T \\
&\approx S_0 \left(1 \pm \text{IV}\sqrt{T}\right) && e^x \approx 1 + x \text{ for small } x
\end{aligned}
$$

Here μ_g is the mean growth rate, Z is a standard normal random variable, and $S_\pm$ is the share price one standard deviation above or below S_0. For example, a share at S_0 = 100 USD/share with IV = 40% and 30 days to expiration (T = 30/365 years) has IV$\sqrt{T}$ = 0.115, so the range is about 88.5 to 111.5 USD/share. Under the model, the share price ends inside this range about 68% of the time.

Decision-Rule for American Options

At each node, the American holder decides to exercise now or hold for the future.



The Early-Exercise Premium

At expiration, each node's value is the contract's payoff. Working backward, the American holder takes the larger of exercising now and holding for the future, while the European holder can only hold:

$$\hat{V}^{\mathsf{AM}}_{t,k} = \max\Big\{\underbrace{V(K, S_{t,k})}_{\text{exercise now}},\ \underbrace{\mathcal{D}^{-1}_{t+1,t}(g_y)\cdot\left(q\,\hat{V}^{\mathsf{AM}}_{t+1,k+1} + (1-q)\,\hat{V}^{\mathsf{AM}}_{t+1,k}\right)}_{\text{hold for the future}}\Big\}$$

$$\hat{V}^{\mathsf{EU}}_{t,k} = \mathcal{D}^{-1}_{t+1,t}(g_y)\cdot\left(q\,\hat{V}^{\mathsf{EU}}_{t+1,k+1} + (1-q)\,\hat{V}^{\mathsf{EU}}_{t+1,k}\right)$$

Here $\hat{V}_{t,k}$ is the contract's value at the node with k up moves in t steps, $S_{t,k} = S_0 u^k d^{t-k}$ is the share price there, and $\mathcal{D}^{-1}_{t+1,t}(g_y) = e^{-g_y \Delta t}$ discounts one time step. The hold term is the European premium formula applied over one step. The premium is $\mathcal{P} = \hat{V}_{0,0}$, and $\mathcal{P}^{\mathsf{AM}} - \mathcal{P}^{\mathsf{EU}} \geq 0$ is the early-exercise premium.

Pseudocode: Pricing on the CRR Lattice

Initialize: given S_0, K, the payoff function $V(K, S)$ of the call or put, T, σ, g_y, and the number of time steps h. Compute $\Delta t = T/h$, u and d, q, and $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

1. Build the share price lattice: $S_{t,k} \leftarrow S_0 u^k d^{t-k}$ for each level $t = 0, 1, \ldots, h$ and node $k = 0, 1, \ldots, t$.
2. At expiration, set each node value to the payoff: $\hat{V}_{h,k} \leftarrow V(K, S_{h,k})$ for $k = 0, 1, \ldots, h$.
3. For each level $t = h-1, h-2, \ldots, 0$ **do**:
   - For each node $k = 0, 1, \ldots, t$ **do**:
     a. Compute the value of holding for the future: $H_{t,k} \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot (q\,\hat{V}_{t+1,k+1} + (1-q)\,\hat{V}_{t+1,k})$.
     b. If the contract is American, **then**: $\hat{V}_{t,k} \leftarrow \max\{V(K, S_{t,k}),\ H_{t,k}\}$. Otherwise, $\hat{V}_{t,k} \leftarrow H_{t,k}$.
4. The premium is the value at the root: $\mathcal{P} \leftarrow \hat{V}_{0,0}$.

A Hypothetical Example from Hull

Consider a put contract.

- Contract: the strike price is $K = 52$ USD/share on a stock whose current price is $S_0 = 50$ USD/share.
- Lattice: two time steps of 1 year each. In each period, the stock price moves up by a factor of $u = 1.2$ or down by a factor of $d = 0.8$.
- Pricing inputs: the continuously compounded risk-free rate is $g_y = 5\%$, so the risk-neutral probability of an up move is $q = (e^{0.05} - 0.8)/(1.2 - 0.8) = 0.6282$.

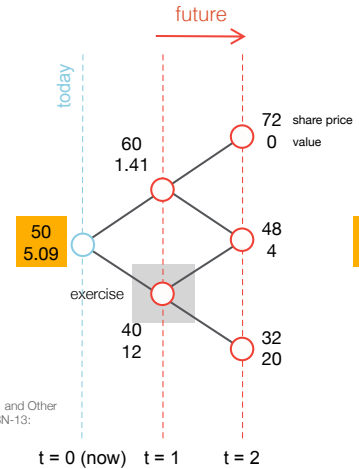
What is the price of the American and European put option contracts?

Source: Chapter 13 of Hull, Options, Futures, and Other Derivatives, 9th edition (2015).

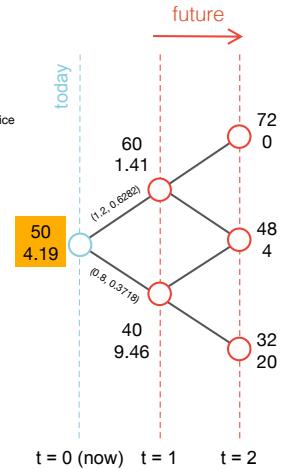
American versus European Put on One Lattice

Example: Hull 13.4 and 13.5

American Put contract



European Put contract



Reference: John C. Hull "Options, Futures, and Other Derivatives" 9th edition, 2015. Pearson, ISBN-13: 978-0133456318

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Early Exercise in the Hull Example

At the down node at $t = 1$, exercising pays 12 USD/share, while holding is worth 9.46 USD/share, so the American holder exercises.

That one decision raises the premium from 4.19 to 5.09 USD/share, an early-exercise premium of about 0.90 USD/share.

Review on your own: CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb

Compute the Price of an American Call and Put Option Contract

Now that we've explored the decision rule through the Hull example, let's apply these concepts to real market data: CRR prices for American calls and puts from the course's options archive, compared with market prices across strike.

When does early exercise pay? For a call on a share that pays no dividend, exercising early never helps. Exercising early means paying the strike sooner and owning the shares if the price then falls below the strike, so $\mathcal{P}_c^{\text{AM}} = \mathcal{P}_c^{\text{EU}}$.

For a put, exercising early can pay when the put is deep in the money. The holder receives the strike now and earns the risk-free rate on it, so $\mathcal{P}_p^{\text{AM}} > \mathcal{P}_p^{\text{EU}}$ is possible.

Example 1: CHEME-5660-L9b-Example-AmericanContractsPremium-Fall-2026.ipynb

Monte Carlo Pricing of American Options

Previously, we priced European contracts by simulating share prices under $\mathbb{Q}$ and averaging the discounted payoffs. Longstaff and Schwartz extended this approach to American contracts.

The decision rule needs the expected value of holding, but each path has only one future. Longstaff and Schwartz estimate it by regression across paths, working backward from expiration:

Initialize: given S_0, K, the payoff function $V(K, S)$ of the call or put, T, σ, g_y, the number of exercise dates N after today, and the number of paths M. Compute $\Delta t = T/N$ and $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

Reference: Longstaff, F. A., and Schwartz, E. S. (2001). Valuing American options by simulation: a simple least-squares approach. Review of Financial Studies 14(1): 113–147.

Pseudocode: Longstaff–Schwartz

1. Simulate: generate M share price paths under $\mathbb{Q}$ on the exercise dates $t = 0, 1, \ldots, N$, where $S_i(t)$ is the share price on path i at date t.
2. At expiration, set each path's cash flow to its payoff: $Y_i \leftarrow V(K, S_i(N))$ for $i = 1, 2, \ldots, M$.
3. For each date $t = N-1, N-2, \ldots, 1$ **do**:
   a. Discount each cash flow back one date: $Y_i \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot Y_i$.
   b. Regress: on the in-the-money paths, fit $\hat{C}(S) = \beta_0 + \beta_1 S + \beta_2 S^2$ to the pairs $(S_i(t), Y_i)$ by least squares.
   c. Decide: for each in-the-money path, if $V(K, S_i(t)) > \hat{C}(S_i(t))$, **then**: exercise, so $Y_i \leftarrow V(K, S_i(t))$.
4. Discount the cash flows to today and average them: $\bar{Y} \leftarrow \mathcal{D}_{1,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} Y_i$.
5. The premium is $\mathcal{P}^{\text{AM}} \leftarrow \max\{V(K, S_0),\ \bar{Y}\}$.

The Early-Exercise Premium by Simulation

Let’s price an American put by simulation and compare it with the lattice and the European premium.

Holding on every path gives the European Monte Carlo premium, so the difference estimates the early-exercise premium.

Example 2: CHEME-5660-L9b-Example-LSM-Premium-Fall-2026.ipynb

Optional Advanced Material

CRR factors: advanced/crr_factors/
CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

In this notebook, we derive the up and down factors of the CRR model by matching the variance of the lattice's one-step log return to the variance of geometric Brownian motion. We then check that the match holds under the risk-neutral probability, and find how small the time step must be to keep that probability between zero and one.

Notebooks: lectures/week-9/L9b/advanced, also linked from the lecture notebook.

Summary

Today we priced American options with the CRR binomial model and Monte Carlo simulation.

Key takeaways

- Binomial trees price American options: at each node, the holder exercises or holds, so backward induction computes the premium.
- Early exercise can pay for puts: the American premium exceeds the European premium only when exercising early pays at some node. That never happens for a call on a share with no dividend.
- Simulation estimates the same premium: Longstaff–Schwartz works backward along simulated paths, regressing the value of holding on the share price. Holding on every path gives the European premium.

The early-exercise premium is what the right to exercise before expiration is worth, and the lattice and simulation give two ways to compute it.